INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II ,Third Semester Examination, November - 2024

Subject Code: BCACC301

Subject Name: Data Structures with C

Full Marks: 30 Duration: 1 Hour

Part - A
Attempt 5 questions
Each question carries 2 marks (2 × 5)

1. Describe Linked list in brief. 2

   or

   Differentiate between array and linked list? 2

2. Define Node in linked list. 2

   or

   Define circular linked list. 2

3. State the advantages of circular lists over doubly linked list. 2

   or

   Differentiate between preorder and inorder traversal of a tree. 2

4. Define Height of tree. 2

   or

   What are the two ways of implementing a tree? 2

5. List the operation to be performed on a binary tree. 2

   or

   List the types of linked lists. 2

Part - B
Attempt 2 questions
Each question carries 5 marks (5 × 2)

6. Explain the advantages and disadvantages of using a doubly linked list over a singly linked list. 5

   or

   Explain the process of inserting a node at the beginning, and end of a singly linked list. 5

7. Describe the steps involved in reversing a singly linked list, and write the function for it. 5

or

Construct an expression tree for the expression (a+b*c) + ((d*e+f)*g). Give the outputs while applying pre-order traversals. 5

Part - C

Attempt 1 question

Each question carries 10 marks (10 × 1)

8. Create a binary search tree for the following numbers start from an empty binary search tree. 45,26,10,60,70,30,40 Delete keys 10, 60 and 45 one after the other and show the trees at each stage. 10

or

Develop an algorithm or pseudo code to implement Bubble Sort. Explain how it works and why it is considered inefficient for large datasets. 10

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